Merge Sort:

```
def merge_sort(arr):
    # Base case: a list with 0 or 1 element is already sorted
    if len(arr) <= 1:
        return arr

    # Divide: find the middle point and split the list into two halves
    mid = len(arr) // 2
    left_half = arr[:mid]
    right_half = arr[mid:]

    # Conquer: recursively sort both halves
    left_half = merge_sort(left_half)
    right_half = merge_sort(right_half)

    # Combine: merge the sorted halves
    return merge(left_half, right_half)

def merge(left, right):
    merged_list = []
    i = 0 # pointer for the left list
    j = 0 # pointer for the right list

    # Compare elements from both lists and add the smaller one to the result
    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            merged_list.append(left[i])
            i += 1
        else:
            merged_list.append(right[j])
```

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```
        j += 1

    # If one list is exhausted, add the remaining elements from the other list
    while i < len(left):
        merged_list.append(left[i])
        i += 1

    while j < len(right):
        merged_list.append(right[j])
        j += 1

    return merged_list

# Driver program to test the code
if __name__ == '__main__':
    my_list = [12, 11, 13, 5, 6, 7]
    print(f"Given list is: {my_list}")
    sorted_list = merge_sort(my_list)
    print(f"Sorted list is: {sorted_list}")
```

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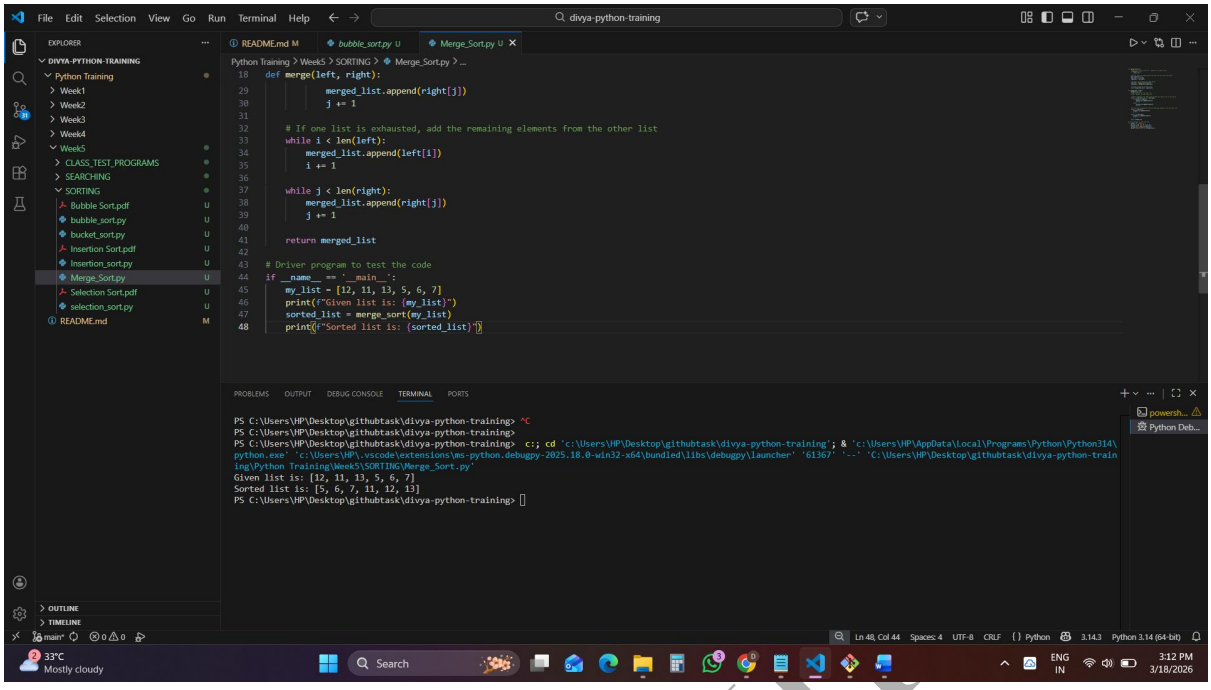
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bubble_sort.py U
Merge_Sort.py U
Python Training > Week5 > SORTING > Merge_Sort.py > ...
def merge(left, right):
merged_list.append(right[j])
j += 1
# If one list is exhausted, add the remaining elements from the other list
while i < len(left):
merged_list.append(left[i])
i += 1
while j < len(right):
merged_list.append(right[j])
j += 1
return merged_list
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PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\HP\Desktop\githubtask\divya-python-training> ^C
PS C:\Users\HP\Desktop\githubtask\divya-python-training>
PS C:\Users\HP\Desktop\githubtask\divya-python-training> c:; cd 'c:\Users\HP\Desktop\githubtask\divya-python-training'; & 'c:\Users\HP\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\HP\.vscode\extensions\ms-python.debugpy-2025.18.0-win32-x64\bundled\libs\debugpy\launcher' '61367' '--' 'C:\Users\HP\Desktop\githubtask\divya-python-training\Python Training\Week5\SORTING\Merge_Sort.py'
Given list is: [12, 11, 13, 5, 6, 7]
Sorted list is: [5, 6, 7, 11, 12, 13]
PS C:\Users\HP\Desktop\githubtask\divya-python-training>
Ln 48, Col 44 Spaces: 4 UTF-8 CRLF {} Python 3.14.3 Python 3.14 (64-bit)
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